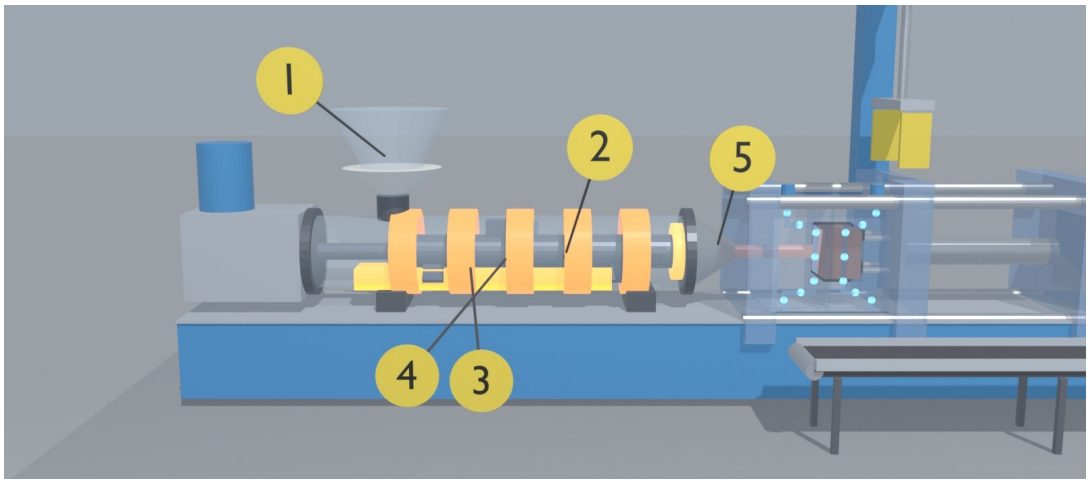
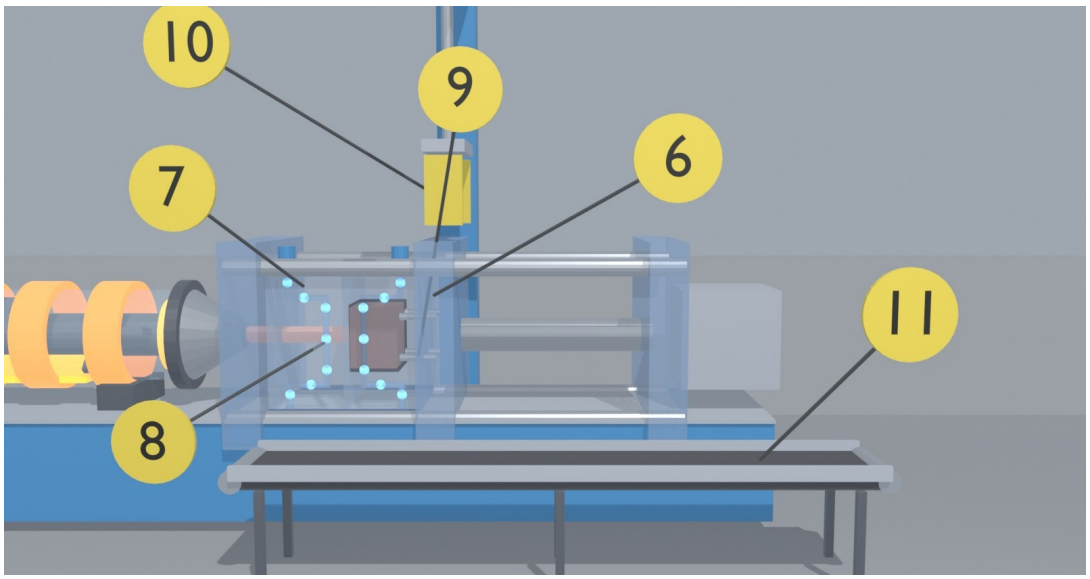
Injection molding: parts and functions

Look at the two pictures of the injection molding machine. The left picture shows the injection unit with the hopper, the barrel, the screw, the heater bands and the nozzle (parts 1 to 5). The right picture shows the clamping unit with the mold, the ejector pins, the take-out robot and the conveyor (parts 6 to 11). Each picture has different numbers. The numbers show the parts in the order the process uses them. Match the names in the word box to the numbers. Then write in your own words what each part does.





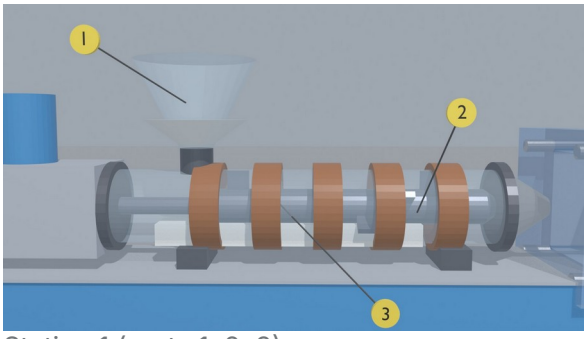
Cooling channels • Screw • Conveyor • Hopper • Clamping unit • Ejector pins • Barrel • Take-out robot • Heater bands • Mold • Nozzle

Parts and functions

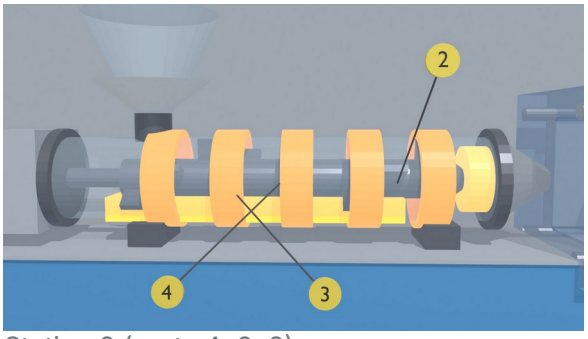
No.	Name of the part	What does it do?
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

Pictures of the six stations

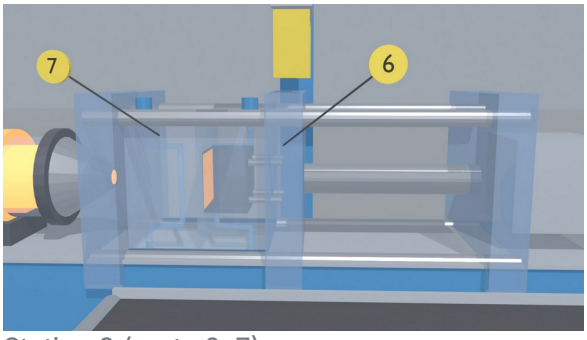
Each picture shows one step of the process. Use them to find the numbered parts. The numbers are the same as in the table on page 2.



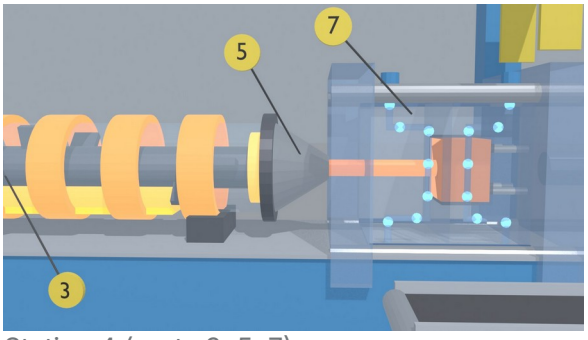
Station 1 (parts 1, 2, 3)



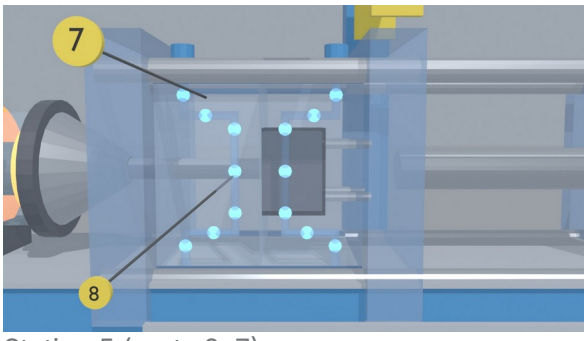
Station 2 (parts 4, 2, 3)



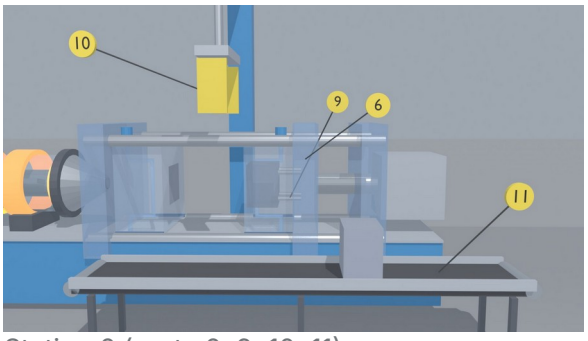
Station 3 (parts 6, 7)



Station 4 (parts 3, 5, 7)



Station 5 (parts 8, 7)



Station 6 (parts 6, 9, 10, 11)